Name _____ Student ID _____

Part I: Multiple Choice (20 points)

Circle the one correct answer in each question.

1. What is a trap in an operating system?

   (a) It is a mechanism that transfers control to an OS handler.
   (b) It lets a user process capture another process's output.
   (c) It is a system call that captures user input.
   (d) It is a way to capture bugs.
   (e) There is no trap mechanism in an operating system.

2. What protection risk arises if user mode and kernel mode are not separated?

   (a) A process cannot be preempted by the operating system.
   (b) The operating system cannot access disks or external devices.
   (c) A process may access another process's memory.
   (d) The operating system can prevent a process from directly accessing devices.
   (e) Creating child processes becomes impossible.

3. Which set gives the four components of a process in the model used in class?

   (a) Program and data only
   (b) Program, data, and control state
   (c) Program, data, and process state
   (d) Data, control state, and process state
   (e) Program, data, control state, and process state

4. Which OS goal is specifically intended to protect one application from another?

   (a) Performance
   (b) Isolation
   (c) Reliability
   (d) Mobility
   (e) All of these are issues for protecting applications from one another.

5. Which low-level hardware mechanism is needed in a preemptive system to switch a CPU-bound process?

   (a) MMU
   (b) Timer interrupt
   (c) Cache
   (d) TLB
   (e) None of these.

6. Which part of a process is stored in kernel space?

   (a) Text segment
   (b) Data segment
   (c) Stack segment
   (d) Process descriptor
   (e) All four items are outside kernel space.

7. In a conventional process address space, which region grows in the opposite direction from the heap?

   (a) Text
   (b) Heap
   (c) Stack
   (d) Program
   (e) They are all growing in the same direction.

8. When would a process in the ready state be moved directly to the blocked state?

   (a) When it reads a file.
   (b) When it reads from memory.
   (c) When a timer interrupt occurs.
   (d) It cannot happen.
   (e) Both when it reads from a file and from memory.

9. Which event can move a process into the ready state?

   (a) A timer interrupt occurs.
   (b) An I/O transfer is completed.
   (c) A process is created.
   (d) All of the events in (a), (b), and (c).
   (e) None of the events in (a), (b), and (c).

10. What does it mean for a process to be a zombie?

    (a) It is running on a processor and cannot be kicked off.
    (b) It is waiting for I/O indefinitely.
    (c) It is ready but has never been selected to run.
    (d) It has terminated but remains in the process table until its exit status is collected.
    (e) There is no such thing as a zombie in an operating system.

11. Which process-control block stores a process's state and saved CPU context?

    (a) PCB
    (b) TLB
    (c) Stack
    (d) MMU
    (e) Queue

12. Which situation can leave a zombie process in the system?

    (a) The parent remains alive after the child exits and does not collect its exit status.
    (b) The parent creates more than one child.
    (c) The child also creates a child process.
    (d) The parent waits for the child and collects its exit status.
    (e) The child executes another program.

13. If a parent successfully calls wait() before its own print statement, what is guaranteed?

    (a) The child always prints after the parent.
    (b) A zombie process is always created.
    (c) The child starts again from the beginning of main().
    (d) A segmentation fault always occurs.
    (e) The parent's print statement runs only after the child terminates.

14. The following C program:

```
#include <stdio.h>
#include <unistd.h>

int main(void) {
    fork();
    fork();
    printf("hello");
    return 0;
}
```

    Assume both calls succeed. How many times will the word “hello” be printed?

    (a) Only once
    (b) Twice
    (c) Three times
    (d) Four times
    (e) Eight times

15. Which events can cause a running process to give up the CPU?

    (a) A timer interrupt.
    (b) Calling a blocking I/O transfer.
    (c) Calling wait() when it must block.
    (d) Calling yield().
    (e) All of the events in (a), (b), (c), and (d).

16. Which statement is NOT true about MLFQ?

    (a) It has multiple queues at different priority levels.
    (b) A new job enters at the lowest priority level.
    (c) A ready job is on exactly one queue.
    (d) Round robin is used among jobs in the same queue.
    (e) A periodic priority boost can move jobs to the highest-priority queue.

17. In the C storage-class model covered in class, which class requests register storage for a local variable?

    (a) Auto
    (b) Register
    (c) Extern
    (d) Static
    (e) All storage classes are executed at the same speed by the CPU.

18. Which statement about partition allocation is true?

    (a) Single-partition allocation is more efficient than multiple-partition allocation in terms of memory usage.
    (b) Flexible-size partitions do not suffer from external fragmentation.
    (c) Fixed-size partitions suffer from internal fragmentation.
    (d) To solve internal fragmentation, the OS needs to do memory coalescing and compaction.
    (e) None of these statements is true.

19. What does the short-term process scheduler do?

    (a) Selects a program to load into memory.
    (b) Selects a process to run.
    (c) Selects a process to remove from memory.
    (d) Selects memory to assign to a process.
    (e) Selects a process to put into memory.

20. What does a page-replacement algorithm do when a frame is needed?

    (a) Selects a page to load into memory.
    (b) Selects a page to run.
    (c) Selects a page to remove from memory.
    (d) Selects memory to assign to a page.
    (e) Selects a page to replace a process.

21. What is not used in address translation in any of the schemes covered in class?

    (a) MMU
    (b) Page table
    (c) TLB
    (d) Segment table
    (e) All of them may be used in address translation, depending on the scheme.

22. What is a disadvantage of segmentation when each segment occupies contiguous physical memory?

    (a) Address translation from virtual to physical addresses is impossible.
    (b) It cannot protect one process's memory from another process.
    (c) It can suffer from external fragmentation.
    (d) It always requires a large table for address translation.
    (e) It has no disadvantages.

23. What does demand paging mean?

    (a) A page is loaded only when it is referenced and is not already resident.
    (b) A page is always in memory for on-demand access.
    (c) A page is pre-loaded on user demand.
    (d) A page is pre-loaded based on the probability of its use.
    (e) There is no such thing as demand paging in an operating system.

24. Which step does NOT occur while handling a valid page fault?

    (a) The OS saves the faulting process's context.
    (b) If no free frame is available, the OS selects a victim page and reuses its frame.
    (c) The OS locates the requested page and reads it from disk.
    (d) The OS updates the page-table entry to mark the page present.
    (e) All of these steps may occur.

25. What does each segment-table entry contain in the scheme shown in Lecture 5?

    (a) A base and a bound.
    (b) A direction and a valid bit.
    (c) A base and a direction.
    (d) A bound and a valid bit.
    (e) All four items in (a)--(d).

26. With a fixed virtual address-space size, what is one effect of using a larger page size?

    (a) More problems with external fragmentation.
    (b) Smaller TLB reach.
    (c) A smaller page table.
    (d) More processes can be created.
    (e) There is no effect from the page size.

27. Which single-queue policy rotates among all ready jobs, giving each a time slice in turn?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLFQ
    (e) All of the schedulers have the same fairness.

28. Which scheduler does not need a timer interrupt for time-slice preemption?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLFQ
    (e) All of the schedulers have the same fairness.

29. Where does virtual memory save the image of memory?

    (a) In the virtual address space.
    (b) On a virtual cloud.
    (c) On a virtual private network.
    (d) In the kernel of the operating system.
    (e) On secondary storage.

30. What is the response time of a process?

    (a) Time from when a process starts running until it finishes.
    (b) Time from when a process enters the system until it finishes.
    (c) Time from when a process enters the system until it starts running.
    (d) Time from when a process runs until it requests I/O.
    (e) Time a process runs during each cycle.

31. What is the turnaround time of a process?

    (a) Time from when a process starts running until it finishes.
    (b) Time from when a process enters the system until it finishes.
    (c) Time from when a process enters the system until it starts running.
    (d) Time from when a process runs until it requests I/O.
    (e) Time a process runs during each cycle.

32. What operation may be needed to address external fragmentation from flexible-size partitions?

    (a) It limits the degree of multitasking.
    (b) Internal fragmentation.
    (c) Address translation.
    (d) Compaction.
    (e) There is no problem.

33. With a single-level page table and fixed page size, what is a cost of a larger virtual address space?

    (a) More memory is needed to store the page table.
    (b) Internal fragmentation.
    (c) External fragmentation increases.
    (d) TLB reach is smaller.
    (e) There is no problem caused by a larger address space.

34. How can a multilevel page table save memory for a sparse address space?

    (a) The table size is reduced.
    (b) The address space is reduced.
    (c) The page size is increased.
    (d) Every second-level page table must be allocated.
    (e) The OS can omit lower-level page tables for unused regions.

35. In the simplified state model, which state does a newly created process enter after admission?

    (a) Ready
    (b) Running
    (c) Blocked
    (d) Terminated
    (e) Empty

36. Which state is a process in while it is executing on the CPU?

    (a) Ready
    (b) Running
    (c) Blocked
    (d) Terminated
    (e) Empty

37. What fairness risk can SJF create if shorter jobs keep arriving?

    (a) Suffer from convoy effect.
    (b) Highest average waiting time.
    (c) Highest average response time.
    (d) Highest average turnaround time.
    (e) Starvation.

38. Which scheduling metric is Round Robin primarily designed to improve?

    (a) Lowest average turnaround time.
    (b) Lowest average waiting time.
    (c) Starvation could occur.
    (d) Lowest average response time.
    (e) Highest throughput.

39. Which statement about paging is correct?

    (a) A page and a page frame can be different in size.
    (b) The smaller the page size is, the smaller the page table is.
    (c) A larger page transfer is as fast as a smaller page transfer.
    (d) A page is in logical memory; a page frame is in physical memory.
    (e) Demand paging is when a program can demand preloading of a page.

40. Which page-replacement policy is an ideal benchmark but cannot be implemented exactly without knowing future references?

    (a) OPT
    (b) FIFO
    (c) LRU
    (d) LFU
    (e) All of these policies can be implemented exactly.

41. When does a page fault occur?

    (a) A process references a valid virtual page whose present bit is 0.
    (b) A process accesses an address outside its virtual address space.
    (c) A process accesses a page that is already present in physical memory.
    (d) A process accesses a present page that has been modified.
    (e) All of the above cause a page fault.

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Part II: Short Answer (10 points)

Answer all questions briefly in the spaces provided. (1 point each.)

1. Write the full names of MLFQ and STCF.

2. What is the difference between a process and a program?

3. Expand MMU and state its address-translation function.

4. Give two examples of CPU state saved during a context switch.

5. Give two types of information stored in a PCB, other than CPU register values.

6. What is an exited child process called before its living parent reaps its exit status?

7. Give one example each of an I/O-intensive task and a CPU-intensive task.

8. What is thrashing in a virtual-memory system?

9. With base = 16 KiB and bound = 4 KiB, is virtual byte address 4200 valid? Why?

10. Expand TLB and explain why a TLB miss need not cause a page fault.

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Part III: Written Questions (20 points)

Answer all three questions in the answer booklet. Show your work.

1. **(8 points)** Consider the following set of processes (jobs), where the run time of the process is given in milliseconds (ms):

Process	Run Time	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

   The processes are assumed to arrive in the order P_1, P_2, P_3, P_4, P_5, all at time 0.

   (a) Draw three diagrams illustrating the execution of these processes using FIFO, SJF, and RR scheduling with time slice = 1 ms.

   (b) What is the Turnaround time of each process for each of the schedulers in part (a)?

   (c) What is the Response time of each process for each of the schedulers in part (a)?

   (d) Which of the schedulers results in (i) the minimum average Turnaround time, (ii) the minimum average Response time (taking the average over all processes)?

2. **(4 points)** Explain how a commercial bank handles customer during rush hours with the concept of the Multi-level Feedback Queue (MLFQ), in your opinion. Describe the algorithms and policies used and provide suggestions on how to improve the process with the goal of optimizing the average turnaround time of customers.

3. **(8 points)** Draw three tables show the 3 Frames in physical memory using FIFO, LRU and OPT for page replacement policy (separate table for both policies) and find the Effective Access Time (EAT) of each policies when references strings (list of needed pages) as below:

   7, 0, 1, 3, 0, 3, 2, 4, 2, 7, 0, 3, 2, 1, 2, 0, 1, 4, 0, 1 (20 needed pages)

   Assume that the memory access time is 10 ns and the disk access time is 10 ms.

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